Hierarchy not driven by information processing constraints at scale

1 Introduction

Beginning in the Neolithic period, hierarchical power structures abruptly became the dominant form of social organization across independent regions of both the Old and New Worlds, simultaneous with the adoption of agriculture, greater permanence of settlement, and a steep increase in the scale at which cohesive social groups were formed [1]. Understanding what drove this "hierarchical turn," why it was coincident with these other changes, and the extent to which it was contingent or inevitable has vexed anthropologists since the earliest days of the discipline [2].

The most influential explanation for this phenomenon is Gregory Johnson's theory of "scalar communications stress" [3, 4], outlined in two articles dating from 1978 and 1982. In them, Johnson argued that full coordination requires at least one communication channel between all member pairs, meaning the number of channels scales quadratically with size at a rate given by $n(n-1)/2$. Since Johnson observed that no decision-making unit can efficiently manage more than approximately six subordinates, the resulting information load creates a cognitive and organizational bottleneck. When groups exceed this threshold, they must either fission or reorganize into hierarchical structures that reduce the number of relationships each individual must maintain.

The theory has been widely adopted within anthropology and archaeology, informing analyses of settlement patterns, architectural modularity, and nested organizational structures across Holocene societies [5, 6, 7]. Yet its explanatory power rests on assumptions worthy of reexamination. In particular, Johnson's assumption of full connectivity conflates potential with actual information exchange, and moreover has not withstood empirical findings on social network structure made in the intervening time.

Recent work has begun to reopen inquiry into the hierarchical turn using computational methods and advances in network science [8, 9]. Perret et al. (2020) used an evolutionary opinion formation model to show that groups with skewed influence distributions reached consensus faster than egalitarian ones, with this advantage increasing with size [10]. Tóth-Bozó and Meyer (2025) applied a credibility-weighted DeGroot model to random and scale-free networks, finding that topology is the dominant factor shaping convergence dynamics [11]. Golub and Jackson (2010) established that whether a large society converges to accurate beliefs depends critically on whether any individual is too influential — a condition determined by network structure [12].

Building on and synthesizing these results, this study aims to revisit the question of institutional hierarchy's origin with new tools and a revised paradigm, addressing the "network gap" in the existing literature by modeling coordination as a consensus-formation process on a sparse network and measuring coordination cost as the time required to achieve consensus.

2 Methods

2.1 Consensus Dynamics

I adopt the DeGroot updating framework [13]. N agents are embedded in a network $G = (V, E)$; each agent i holds an initial opinion $x_i^{(0)}$ and influence weight $w_i > 0$. Agents update via $\mathbf{x}^{(t+1)} = W\mathbf{x}^{(t)}$, where W is row-stochastic, constructed by adding self-loops to the adjacency matrix, weighting each column j by w_j, and row-normalizing. Under connectedness and aperiodicity, this processes converges to consensus [12]. Coordination cost is operationalized as the spectral-gap-derived consensus time $T_c^* \approx \frac{\log(1/\delta)}{\log(1/|\lambda_2|)}$ for threshold $\delta = 10^{-6}$, where λ_2 is the second-largest eigenvalue of W in modulus and $\gamma = 1 - |\lambda_2|$ is the spectral gap.

2.2 Experimental Design

Three network topologies are compared, all held to mean degree $C = 6$ to operationalize Johnson's span-of-control limit: (1) a **random regular graph** (degree exactly 6) serving as a structurally neutral baseline; (2) a **Watts-Strogatz small-world network** ($k = 6$, rewiring $p = 0.1$) representing naturally emerging social structure with high clustering and short path lengths [14]; and (3) an **augmented b-ary tree** ($b = 5$) representing top-down hierarchy, with additional leaf-to-leaf edges added with probability decaying by hierarchical distance to satisfy the degree constraint.

Three influence weight distributions are crossed orthogonally with topology: equal ($w_i = 1$), uniform ($w_i \sim \mathcal{U}(1, 10)$), and skewed ($w_i \sim \text{Lognormal}(\mu, \sigma)$). For the hierarchy topology, drawn weights are sort-assigned to nodes by tree depth. Group sizes span approximately 10 to 10,000 across 20 values, with 20 independent trials per condition.

3 Results

Figure 1 compares the scaling behavior of the three social network structures on a double-logarithmic scale for each influence distribution. The scaling exponent α is obtained from the slope of the fitted OLS linear regression on the log-transformed simulation data. Under equal and uniform weights, the random regular graph converges fastest, followed by hierarchy, then small-world. All scaling exponents are sublinear. The introduction of skewed influence weights marks a significant shift: scaling becomes less efficient across all

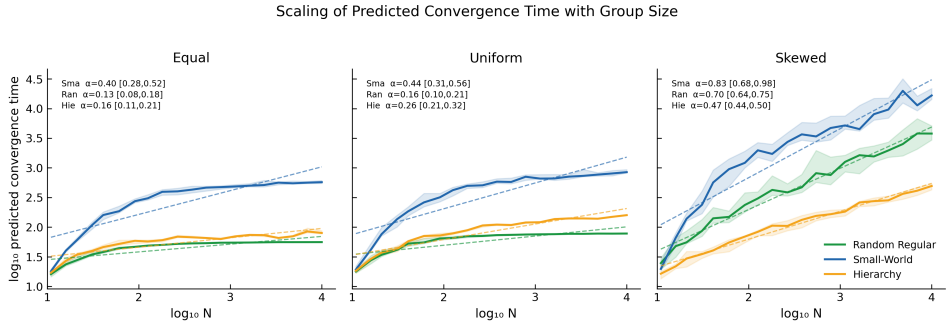

Figure 1: T_c^* *as a function of* $\log N$ *for each model society. Shaded regions denote the interquartile range.*

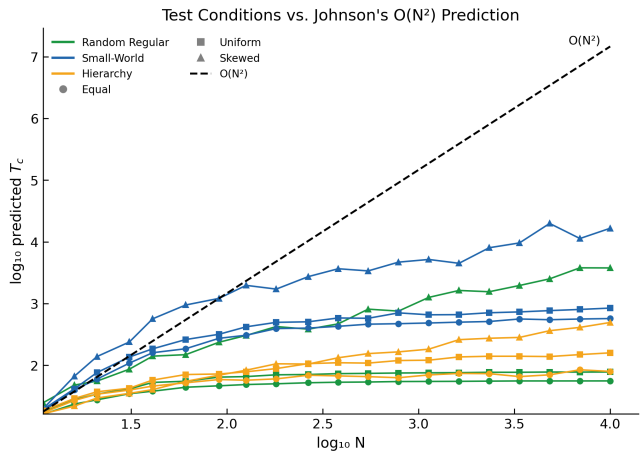

Figure 2: *Single-plot comparison of the scaling efficiency of all tested topology-distribution combinations on a double-logarithmic scale, with the quadratic behavior predicted by scalar stress indicated by a dashed line.*

topologies, and the rank-order of hierarchy and the random graph inverts.

Figure 2 places all nine conditions alongside Johnson's quadratic $O(N^2)$ prediction. At small sizes ($N < 100$), the quadratic curve tracks observed values reasonably well, consistent with the small-group data on which scalar stress theory was grounded. Beyond this range, trajectories diverge sharply: by $N = 10,000$, the Johnson prediction exceeds the nearest observed condition by roughly three orders of magnitude. No tested condition approaches the coordination catastrophe predicted at the scales occupied by chiefdoms and early states.

Figure 3 explores the skewed-influence regime in detail, plotting median predicted $\log T_c$ by topology as a function of the lognormal σ parameter. A crossover is reached at $\sigma \approx 0.69$, beyond which hierarchy exhibits an increasingly pronounced advantage, maintaining low and even decreasing convergence times as heterogeneity grows.

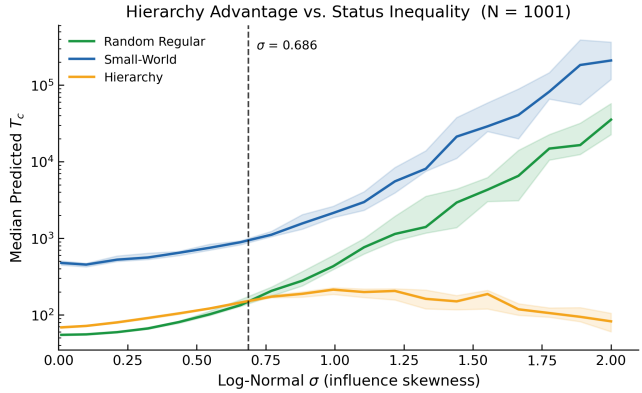

Figure 3: *Median* $\log T_c^*$ *for each topology as a function of the extremeness of status inequality, controlled by the parameter* σ*.*

4 Discussion

It must first be acknowledged that this study does not evaluate scalar stress theory on its own terms, nor is it intended to. Rather, the purpose of this study is to investigate the alternative behaviors that emerge when one upgrades the theoretical foundations with modern tools and knowledge not yet available at the time of the original theory's formulation.

Johnson's framework has been critically evaluated before [15, 16], but perhaps its most significant limitation is an as-yet unarticulated one: namely, that the complete-graph assumption is inconsistent with what is now known about real social networks. Decades of research in network science [14, 17, 18] has established that real social networks are sparse, clustered, and degree-bounded — individuals maintain a fixed number of contacts regardless of group size.

When agents are instead embedded in these more-realistic networks, members are no longer required to exceed their span of control as N grows, and the predicted coordination crisis disappears. None of the nine conditions exhibit poor scaling; consensus time growth is comfortably sublinear, maxing out at $\alpha \approx 0.98$ for the small world society in the skewed-influence regime. This finding is fundamentally incompatible with scalar stress theory as presently conceived, demonstrating that its foundational premise – increasing coordination costs at scale – is altogether absent for sparse networks. If the sparse-network representation is to be accepted, this suggests that constraints on information diffusion at scale are unlikely to be a source of evolutionary pressure toward any particular organizational form.

Even under the hypothesis that societies do optimize for efficient information sharing, a secondary finding is that hierarchy, whether represented by structure or status distribution, does not confer a generic computational advantage. Only topological hierarchy (privileged control over information flow) helps, and only when competing

topologies are disproportionately harmed by influence heterogeneity. The distributional formulation of hierarchy is not only not advantageous, but is actively *dis*advantageous.

Limitations. Several limitations demand acknowledgment. First, the hierarchical topology, augmented with lateral leaf edges to satisfy the mean-degree constraint, may share more structural similarity with the random regular baseline than intended. Second, the log-normal distribution used to generate skewed weights may introduce extreme outliers that disproportionately influence results. Third is the simple fact that networks, while among the best tools we have, are imperfect representations of social systems. True social structure is comprised not of people, but of abstract symbolic roles – and relations between these roles – that are then "filled in" by people [19]. For this reason, I intentionally avoid describing the model as representing social structure directly but rather the communication structure between people, assuming that social structure can be adequately proxied by communication network topology. This may be true but is a strong assumption.

Future directions. There are a number of possible extensions that could strengthen, challenge, and/or refine these findings. Chief among these is model choice. The selection of DeGroot dynamics was motivated by its analytic tractability and established theoretical grounding, but results may not generalize to richer models. Second, higher-order interactions would better represent the broadcast and group-meeting modes common to real-world communication. Third, an optimization approach that allows the network to rewire endogenously in response to coordination pressure could reveal whether the system converges toward hierarchical structure under the right conditions. Finally, comparison with empirical data — particularly archaeological/historical datasets such as Seshat [20] — would allow model predictions to be evaluated against the historical record.

While the results of this study should cast serious doubt on the impact of information-diffusion constraints, they can say nothing (nor are they intended to) about the possibility of scaling pressures rewarding the development of hierarchy via other avenues. It may well be the case, for instance, that it is the potentiality of conflict and disagreement, rather than the cost of communication, that scales quadratically with group size and demands hierarchy as a mechanism for arbitration, in which case the essential logic of scalar stress would remain a valid explanation with only light conceptual revision. It may alternatively be that a third confounding factor is responsible for the relationship between scale and hierarchy, or that a multitude of factors interact in complex ways to produce it, which would demand an entirely new theoretical foundation. This study cannot adjudicate between accounts; only identify a need for the development of novel theories and revisitation of old ones in light of new evidence.

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